$$
\begin{aligned}
&C_{\phi e}^{(12)\,\mathrm{i1\_i2\_}} \to \\
&\frac{1}{16\pi^2}\Bigg(-\frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{2,2,0}\big[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\big]\,\delta_{\mathrm{i1i2}} + \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{3,2,-1}\big[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\big]\,\delta_{\mathrm{i1i2}} - \\
&\quad \frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{2,2,0}\big[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\big]\,\delta_{\mathrm{i1i2}} + \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{3,2,-1}\big[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\big]\,\delta_{\mathrm{i1i2}} + \\
&\quad \frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{3,1,0}\big[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\big]\,\delta_{\mathrm{i1i2}} + \frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{3,2,-1}\big[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\big]\,\delta_{\mathrm{i1i2}} - \\
&\quad \frac{3}{4}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{4,1,-1}\big[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\big]\,\delta_{\mathrm{i1i2}} + \frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{5,1,-2}\big[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\big]\,\delta_{\mathrm{i1i2}} + \\
&\quad \frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{3,1,0}\big[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\big]\,\delta_{\mathrm{i1i2}} + \frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{3,2,-1}\big[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\big]\,\delta_{\mathrm{i1i2}} - \\
&\quad \frac{5}{4}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{4,1,-1}\big[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\big]\,\delta_{\mathrm{i1i2}} + \tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{5,1,-2}\big[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\big]\,\delta_{\mathrm{i1i2}} + \\
&\quad \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{2,2,0}\big[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\big]\,\delta_{\mathrm{i1i2}} - \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{3,2,-1}\big[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\big]\,\delta_{\mathrm{i1i2}} - \\
&\quad \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{3,1,0}\big[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\big]\,\delta_{\mathrm{i1i2}} - \frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{3,2,-1}\big[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\big]\,\delta_{\mathrm{i1i2}} - \\
&\quad \frac{1}{2}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{4,1,-1}\big[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\big]\,\delta_{\mathrm{i1i2}} + \tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{5,1,-2}\big[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\big]\,\delta_{\mathrm{i1i2}} - \\
&\quad \frac{1}{4} m_1\,\tilde{\mu}\, g_1^4\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1]\,\delta_{\mathrm{i1i2}} + \frac{1}{3} m_1\,\tilde{\mu}\, g_1^4\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1]\,\delta_{\mathrm{i1i2}} - \\
&\quad \frac{3}{4} m_2\,\tilde{\mu}\, g_1^2\, g_2^2\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2]\,\delta_{\mathrm{i1i2}} + m_2\,\tilde{\mu}\, g_1^2\, g_2^2\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2]\,\delta_{\mathrm{i1i2}} + \\
&\quad \frac{1}{4} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,1,1,0}\big[m_1, m_{\tilde{e}}^{i1}, m_{\tilde{l}}^{p}\big] - \frac{1}{8} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,2,1,-1}\big[m_1, m_{\tilde{e}}^{i1}, m_{\tilde{l}}^{p}\big] - \\
&\quad \frac{1}{4} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{3,1,1,-1}\big[m_1, m_{\tilde{e}}^{i1}, m_{\tilde{l}}^{p}\big] + \frac{1}{8} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,2,1,-1}\big[m_1, m_{\tilde{l}}^{p}, m_{\tilde{e}}^{i1}\big] + \\
&\quad \frac{1}{4} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,1,1,0}\big[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}\big] - \frac{1}{4} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{3,1,1,-1}\big[m_1, m_{\tilde{l}}^{p}, \tilde{\mu}\big] + \\
&\quad \frac{1}{4} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,2,1,-1}\big[m_1, \tilde{\mu}, m_{\tilde{e}}^{i1}\big] - \frac{3}{4} m_2\,\tilde{\mu}\, g_2^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,1,1,0}\big[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}\big] + \\
&\quad \frac{3}{4} m_2\,\tilde{\mu}\, g_2^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{3,1,1,-1}\big[m_2, m_{\tilde{l}}^{p}, \tilde{\mu}\big] + \frac{3}{4} m_2\,\tilde{\mu}\, g_2^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,2,1,-1}\big[m_2, \tilde{\mu}, m_{\tilde{l}}^{p}\big] - \\
&\quad \frac{1}{2} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,1,1,0}\big[\tilde{\mu}, m_1, m_{\tilde{e}}^{i1}\big] + \frac{1}{2} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{3,1,1,-1}\big[\tilde{\mu}, m_1, m_{\tilde{e}}^{i1}\big] + \\
&\quad \frac{1}{4} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{1,1,1,1,0}\big[m_1, m_{\tilde{e}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}\big] + \frac{1}{8} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\big[m_1, m_{\tilde{e}}^{i1}, m_{\tilde{l}}^{p}, \tilde{\mu}\big] + \frac{1}{4}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, a_e^{pi2}\, \mathrm{LF}_{1,1,1,1,0}\big[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}, \tilde{\mu}\big] - \\
&\quad \frac{1}{8}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, a_e^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\big[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}, \tilde{\mu}\big] - \frac{1}{4}\tilde{\mu}\, \overline{y_e}^{pi1}\, \overline{y_e}^{rs}\, y_e^{ri2}\, a_e^{ps} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde{e}}^{s}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{r}, \tilde{\mu}\big] + \frac{1}{8} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde{e}}^{i1}, m_1, m_{\tilde{l}}^{p}, \tilde{\mu}\big] - \\
&\quad \frac{1}{8}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, a_e^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{p}, \tilde{\mu}\big] + \frac{1}{2}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, a_e^{pi2} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{i1}, m_{\tilde{e}}^{i2}\big] - \frac{1}{8} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{i1}, \tilde{\mu}\big] + \\
&\quad \frac{1}{8}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, a_e^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde{l}}^{p}, m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}\big] - \frac{1}{8} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, y_e^{pi2} \\
&\qquad \mathrm{LF}_{2,1,1,1,-1}\big[\tilde{\mu}, m_1, m_{\tilde{e}}^{i1}, m_{\tilde{l}}^{p}\big] + \frac{1}{8}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pi1}\, a_e^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\big[\tilde{\mu}, m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{p}\big]\Bigg)
\end{aligned}
$$